

ENGINE

EQUIP NUM: 02

SERIAL NUMBER: 4FD05213

CAT G3406



No Action Required

Interp By: Salsally Ramadhani

Interpreted On: 22-May-26

All Test Results Appear Acceptable. Take Oil Samples At 250 Hour Intervals To Monitor Condition.

Semua Hasil Tes Dalam Batas Normal. Ambil Sampel Oli Pada Interval 250 Jam Untuk Memonitor Kondisi.

SAMPLE INFORMATION

	08-May-26	24-Sep-25	25-Jun-24	25-May-23
Sampled Date	08-May-26	24-Sep-25	25-Jun-24	25-May-23
Sample Id	J21A-56141-0215	J21A-55270-1720	J21A-54181-0517	J21A-53152-0321
Lab Date	21-May-26	27-Sep-25	29-Jun-24	01-Jun-23
Meter [Hr]	35235	30281	30000	21704.0
Comp Meter [Hr]	35235	30281	30000	
Meter On Fluid	0	0	0	909.0
Fluid Brand	PERTAMINA	PERTAMINA	PERTAMINA	PERTAMINA
Fluid Weight	40	40		40
Fluid Type	NG LUBE	NG LUBE	NG LUBE	NG LUBE
Fluid Change	Y	N	Y	Y
Filter Change	N	N	Y	Y
Make Up Fluid [L]	110			
Kidney Loop	U	U		
Total Fluid Added	110	0	0	0

WEAR METAL/CONTAMINANT/ADDITIVES

	08-May-26	24-Sep-25	25-Jun-24	25-May-23
ELEMENTS (ppm)				
Cr Chromium	0	0	1	0
Pb Lead	0	0	1	0
Fe Iron	3	1	6	1
Cu Copper	10	0	350	2
Al Aluminum	1	1	1	1
Sn Tin	0	0	0	0
Ni Nickel	0	0	0	0
B Boron	1	2	0	0
K Potassium	0	0	0	1
Na Sodium	1	0	0	1
Si Silicon	1	4	4	2
Ca Calcium	1378	1363	1365	1359
P Phosphorus	259	246	286	247
Zn Zinc	364	302	322	322
Mg Magnesium	45	4	4	6
Mo Molybdenum	2	1	1	0

For additional sample history, go to:

[S.O.S WEB](#)

OIL CONDITION AND PHYSICAL/CHEMICAL TEST

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
OIL CONDITION (JOAP)					
ST Soot		0	0	0	0
OXI Oxidation		11	9	9	7
NIT Nitration		6	4	5	3
SUL Sulfate By-Product		16	16	15	13
Sulfur Products		16	16	15	13

FUEL CONTAMINATION

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
F Fuel		N	N	N	N

VISCOSITY AT 100°C (Centistokes)

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
V100 Viscosity at 100 C		13.52	13.56	13.27	13.50

TOTAL BASE NUMBER (mg KOH/ g)

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
TBN Total Base Number		3.2	4.9	3.9	4.2

TOTAL ACID NUMBER (mg KOH/g)

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
TAN Total Acid Number		1.58			

WATER CONTAMINATION

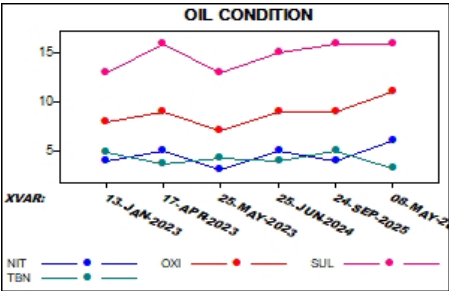
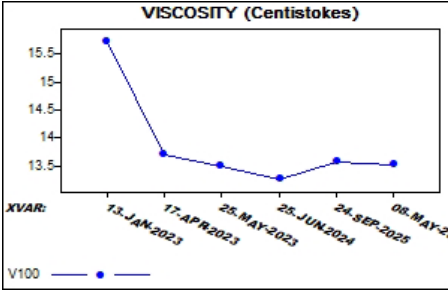
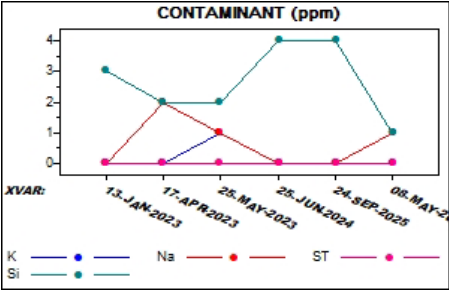
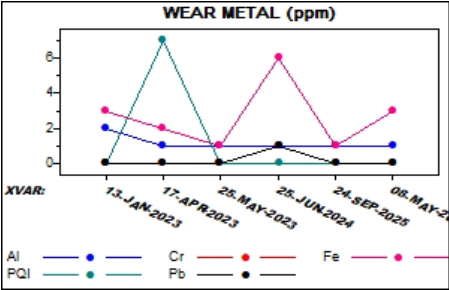
		08-May-26	24-Sep-25	25-Jun-24	25-May-23
W Water		N	N	N	N

PARTICLE QUANTIFIER INDEX

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
PQI PQ Index		0	0	0	0

PARTICLE COUNT AND OTHER ANALYSIS

		08-May-26	24-Sep-25	25-Jun-24	25-May-23
VISUAL					
PCV PC Visual		-	-	-	-



Result Explanation Section

Element – TU.SOS.WIN.MP.02.R16
ppm (mg/Kg) = Parts Per Million

Wear Metals

Cr = Chromium
Pb = Lead
Fe = Iron
Cu = Copper
Al = Aluminium
Sn = Tin
Ni = Nickel

Contaminant

B = Boron
K = Potassium
Na = Sodium
Si = Silicon

Additives

Ca = Calcium
P = Phosphorus
Zn = Zink
Mg = Magnesium
Mo = Molybdenum

Particle Count – ASTM D7647-24

Count / mL
4µ = Particle Count at 4 micron
6µ = Particle Count at 6 micron
14µ = Particle Count at 14 micron

Particle Count – ISO 4406-21*
ISO CODE

Visual*

Dark (DRK) = Dark & Dirty Fluid
Cloudy (CLO) = Milky or Cloudy Fluid
Clear (CLR) = Clear Visible Particle
Vol = Volume
HBL = High Baseline
MetF = Metallic Reactive
MetY = Metallic Non-Reactive Yellow
MetW = Metallic Non-Reactive White/Gray
Nmet = Non-Metallic
LQ = Low Quantify
MQ = Medium Quantify
HQ = High Quantify

Water Criteria

Water Content (% Wt) ASTM D6304-20*
TU.SOS.WIN.MP.09.R7
N = Negatif (<0.1%)
T = Trace (0.1% - <0.5%)
P = Positive (0.5% - 1.0%)
E = Excessive (>1.0%)

Fuel Criteria

Fuel Content (% Vol) – ASTM D7593-22*
N = Negative
P = Positive

Oil Condition (UFM) – ASTM E2412-23a

ST = Soot
OXI = Oxidation
NIT = Nitration
SUL = Sulfation

Particle Quantifier (Index) – ASTM D8184-18e1
Viscosity 100°C / Viscosity 40 °C (cSt) – ASTM D445-24
Total Base Number (mg KOH/g) – ASTM D4739-23
Total Acid Number (mg KOH/g) – ASTM D664-24 (Metode A)*
Chlorine (ppm (mg/L)) – ASTM D7536-24*
Sulfur (ppm (mg/L)) – ASTM D2622-24a*

*Not included in accreditation scope



Approved by

Siti Yuniarti
Sr. Supervisor S O S Laboratory

- This analysis is indicative, based on information from the label and the sample provided. Any decision regarding component improvement is entirely at the customer's discretion.
- The results of this analysis apply only to the sample received and tested by the laboratory and do not represent the entire batch or product.
- The sample was received in a sealed plastic bottle, without physical damage, and without any sampling process conducted by the laboratory.
- It is recommended that further discussion regarding this sample be carried out within 7 days from the date the report is issued.